To Intervene or not to Intervene: Examining Facilitator Tendencies in Online Discussions

Anonymous ACL submission

Abstract

The question of how and when to intervene in online discussions—whether through expert or natural facilitation—is a long-standing concern in the social sciences. This debate has gained new urgency with the advent of Large Language Models, which ostensibly make automated, large-scale intervention increasingly feasible. In this study, we examine human and Large Language Model tendencies of intervention, with a specific focus on positive and negative reinforcement. We then use transformers to [DETAILS TO BE ADDED]. Our results show [DETAILS TO BE ADDED].

1 Introduction

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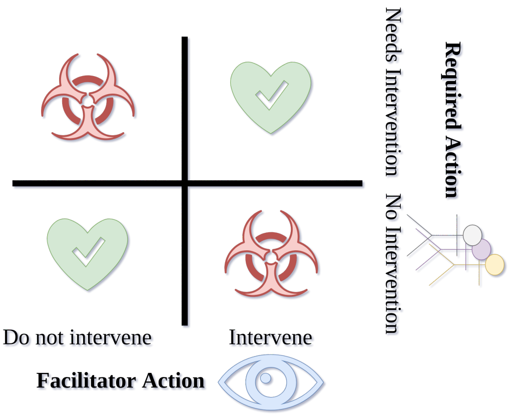

Figure 1: At any point in a discussion, the facilitator must decide whether to intervene or not. Not intervening when the situation calls for it may lead to escalation and derailment. On the other hand, intervening when the discussion is already going well, may lead to irritation and toxicity from participants.

Deciding when to intervene in a discussion is a matter of great importance. Intervening too early or too frequently may lead to users expressing frustration at being constantly interrupted, while intervening too late may lead to a significant escalation and derailment of the discussion (Korre et al., 2025; Gao et al., 2025). However, this task remains underexplored in literature, with current studies using ground-truth labels on which users are facilitators (Chen et al., 2025a; Schroeder et al., 2024; Gao et al., 2025). The task proves harder when considering that a facilitative comment is not necessarily made by dedicated facilitators; since in online environments, facilitative feedback may be primarily provided by the users themselves (Falk et al., 2024).

2 Related Work

2.1 What are facilitative interventions?

Facilitation (or conversational moderation) refers to the active participation of people in order to improve discussions (Korre et al., 2025). Relevant work assumes that facilitation is conducted by specially trained staff (Schroeder et al., 2024; Park et al., 2012), users appointed by the community (Schaffner et al., 2024), or specialized professionals (Chen et al., 2025a). We identify three main definitions for what a facilitative intervention is:

Definition 1 (Escalation) *A facilitator needs to intervene at the precise point in the discussion at which, without intervention, the discussion would derail or escalate.*

Escalation is usually measured as a function of toxicity (Chang and Danescu-Niculescu-Mizil, 2019), or as the point where the participants request arbitration (e.g., escalate the matter to a site moderator in Wikipedia (De Kock and Vlachos, 2021)).

A large amount of work [CITE] is based on this assumption for a few reasons. (1) The labels are easy to acquire; either a ground truth exists (i.e., "escalation" is defined by the website), or can be easily obtained (i.e., toxicity classification is a well-understood and explored automated task).

(2) The existence of a single, largely unambiguous ground-truth is ideal for the training and evaluation of any ML-based classifier. However, recent social science work insists that facilitators help discussions in many ways other than escalation prevention; such as summarizing discussions (Small et al., 2023), managing participation (Chen et al., 2025a; Schroeder et al., 2024), and providing important information for new members (Park et al., 2012).

Definition 2 (Professional facilitation) *An intervention should occur at the point where a professional facilitator would have intervened.*

This definition is simple and intuitive, but can only be used in domains where (1) professional facilitators exist and (2) they can be identified. These conditions exclude the vast majority of online discussions, as very few web-crawled datasets include such information (see Table 1). It is generally accepted that all comments made by a facilitator should be considered facilitative interventions; for instance, all such comments are assumed to follow some social strategy (Chen et al., 2025a; Schroeder et al., 2024).

Definition 3 (Human judgment) *Facilitation should only occur when most people agree that it is necessary.*

While some may argue that professional facilitators have more accurate judgments due to expertise, in reality most facilitation is handled by either volunteers elevated to the position by their community (Schaffner et al., 2024) or by the users themselves in an ad-hoc fashion (Falk et al., 2024). Given that humans generally hold extensive social experience because of sustained social interactions from birth, they innately understand concepts such as when to speak in a discussion (Nonomura and Mori, 2025). Thus, they are a good baseline for machine-enabled facilitation.

An alternative view however suggests that facilitative comments can be (and indeed are most frequently) made by participants themselves (Falk et al., 2024). In this case, this task is no longer about detecting or predicting whether *a facilitator would speak*, but rather the much more difficult task of whether the comment is *facilitative in nature*.

2.2 Facilitation detection

While some recent work (Schroeder et al., 2024; Chen et al., 2025a) uses LLM-as-a-judge models to annotate data in the field of facilitation, the use of such models may not be an obvious choice for the task of facilitation prediction. It has been demonstrated that LLM facilitators can not decide when they should intervene irrespective of instructions and model family/size. Instead, since these models are trained as chatbot assistants, they constantly attempt to speak (Tsirmpas et al., 2025).

Alternatively, facilitation interventions can be seen as a specific case of turn-taking, the process with which humans take turns speaking in a discussion. The goal of turn-taking is typically to minimize idle time in discussions, while letting most participants take part in the discussion. This process is opaque and largely learned after years of social interaction for humans, while only experimental coarse-grained solutions have been devised for automated systems (Nonomura and Mori, 2025). One solution is to use a dedicated smaller model whose only task is to decide whether a larger model should speak (Chen et al., 2025b).[1] No such approach to our knowledge has been used in facilitative settings.

3 Dataset

3.1 Description

Korre et al. (2025) provide a list of datasets relating to discussion facilitation. We combine all of these datasets (Table 1), preprocess them (§A.1) and standardize their data according to a simple, unified schema (§A.2). Relevant statistics can be found in §A.3.

We thus create a multi-domain dataset focused around facilitation, called the "Prosocial and Effective Facilitation in Konversations (PEFK)" dataset. Unlike its predecessors, PEFK is large in size and rich in comment-level information, allowing us to run thorough analysis between similar datasets. In order to help research in the field of facilitation, we openly release PEFK under a CC-BY-SA license.

3.2 Facilitation in real-life discussions

Figure 2 shows the percentage of comments made by facilitators for each of our datasets. In almost all datasets, a third of the comments are made by facilitators. We do not include the UMOD dataset, since its percentage is based on our own thresholding (see A.1).

[1] The authors specifically use LLM finetuning for the smaller model, but provide no explanation for why a traditional transformer-based classifier wouldn't be sufficient for the task.

Name	#Comments	#Discussions	Domain	Format	Fac.
WikiDisputes (2021)	96,320	8,727	Forum	Text	×
WikiTactics (2022)	3,850	213	Forum	Text	✓
WikiConv (2018)	17,806,373	4,486,983	Forum	Text	×
Conversations Gone Awry / CMV II (2019)	40,607	6,691	Forum	Text	×
CeRI (2012)	3,700	132	Forum	Text	✓
User Moderation (UMOD) (2024)	2,000	1,000	Forum	Text	✓
Intelligence Squared 2 (IQ2) (2016)	34,245	108	Debate	Oral	✓
Why How Who (WHoW) (2025a)	25,542	178	Radio / TV	Oral	✓
Fora (2024)	39,438	262	Deliberation	Oral	✓

Table 1: The datasets used in PEFK with accompanying post-processing statistics. We include a large variety of domains, and two different sources; text-only datasets, and transcripts from oral discussions. All datasets include multi-participant discussions, besides UMOD, which contains comment pairs. The **Fac.** column denotes whether information on which user is a facilitator is provided.

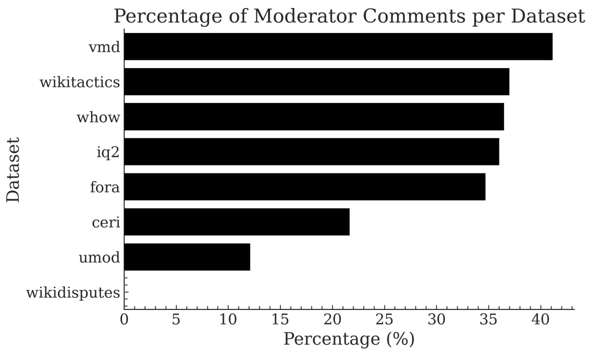

Figure 2: Percentage of moderated comments for each of the sub-datasets of PEFK.

3.3 Text-based vs. transcribed datasets

As the last column of Table 1 would suggest, our data belong in two main sub-modalities; regular written text in the form of posts and comments, and oral transcripts. This reflects a rift in current facilitation research; while some previous work concerns itself with forums and online discussions, others are exclusively focused on live deliberation and debate. Even though we focus on the former, we believe that work from both sides contributes to our understanding of facilitation.

The presence of these two sub-modalities necessitates separate experiments for either of them, since they represent different kinds of data. Posts and comments for instance are self-contained, while transcripts regularly feature interruptions, participants talking in parallel, and linguistic features that are necessary for coordination. They may also feature fundamentally different kinds of facilitation interventions.

4 How to decide when to intervene?

4.1 Do not decide

Dimitris. What most platforms do is to use flagging and reports to decide when to intervene. Mention Grok as a facilitative agent that works by invocation only. Briefly talk about advantages and disadvantages of current approach.

4.2 Use humans

Merits. Korre et al. (2025) state that a fully-automated facilitation is still out of our reach as LLMs tend to be very intruding and repetitive, and fixate on the moderation side of facilitation. Human facilitators, on the other hand, can step in the conversation with a more empathetic outlook, take into deep consideration the context of the conversation, while also being more sensitive to the differences and causes of certain social behaviors.

Disadvantages. However, recruiting human facilitators, which also indicates a sort of expertise, is not a scalable process. In terms of LLM and human annotation alignment, (Movva et al., 2024) focusing on online safety contexts, show that LLMs are quite monodimensional and larger datasets are needed to understand if LLMs can adequately represent different demographic groups. This can also be translated in the context of facilitation, when LLMs *get triggered* only when toxicity appears in the text, disregarding the encouragement of prosocial behavior.

4.2.1 Experimental setup

We should an introductory paragraph that summarizes all the method.

Annotator Profile. We recruited 10 expert annotators for the task of reinforcement detection and classification. All annotators had extensive experience with NLP annotation tasks. The gender distribution was relatively balanced with 4 males and 6 females, while their ages had a median of 25 years. Before the commencement of the annotation process, the annotators completed an agreement form in compliance with GDPR ethical requirements. They were compensated in accordance with Greek legislation.

Annotation Procedure. As described in the previous sections of this article, the goal is to understand *when intervention is needed* as well as *what kind of intervention is needed*. We select a total of 1212 random discussions from all datasets (with a uniform distribution) and select a random section with length 6 comments or less from each. Due to the disagreement with regard to a taxonomy of types of interventions (we should refer the reader to a specific point in the text), we opted for examining only on one type of intervention, that is reinforcement. Since reinforcement [annotation] can be a subjective task (point out to a reference), we adopt an ordinal scale approach, where the annotators rate each dialogue for either type of reinforcement (positive or negative), as well as the possibility of no reinforcement using a scale from 1 to 5. The full annotation guidelines can be found in the Appendix [to add the ref].

4.2.2 Results

Category	Humans	Models	All
Positive	0.34	0.50	0.21
Negative	0.41	0.56	0.28
No-Reinforcement	0.32	0.46	0.20

Table 2: IAA (Fleiss κ) comparison within the human annotator group, model group, and both humans and models.

Apart from some *negligible* exceptions, human annotators very cautious when it comes to intervening, with most often selecting to not intervene (no reinforcement). When it comes to whether they choose positive or negative reinforcement, the discrepancy is not that great, with positive reinforcement being selected slightly more often. However, it must be noted (we need to connect this with the previous results), that despite the slightly higher frequency of positive reinforcement, they agree more on where intervene with negative reinforcement.

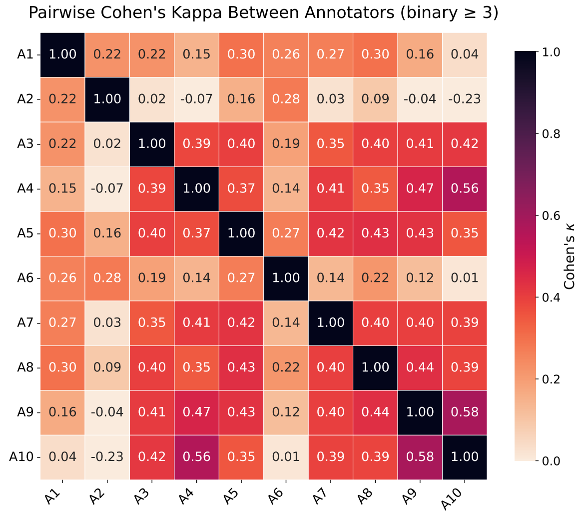

Figure 3: Heatmap of pairwise Cohen's κ scores.

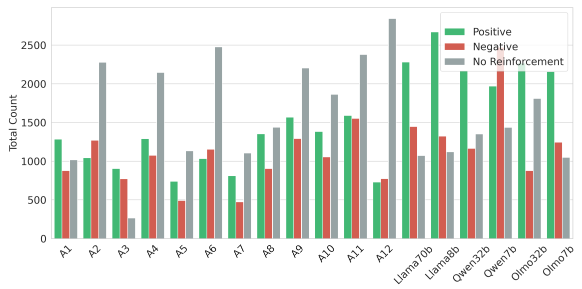

Figure 4: Type of reinforcement the human and LLM annotators choose to intervene with.

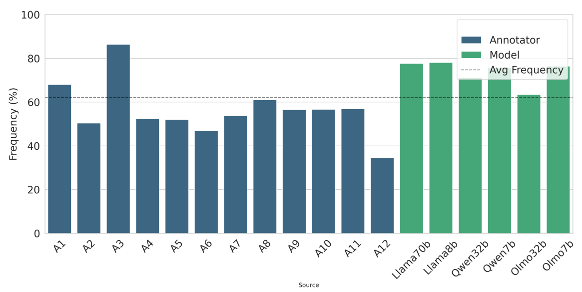

Figure 5: Frequencies of human and LLM annotator interventions.

4.3 Use LLMs

The models on the other hand, are more consistent, with overall higher intervention frequency than you humans (we need to connect this with the fact that models very often become too intrusive). Also, they are more liberal with the intervention types, with the most frequent one being the positive reinforcement, and last on the no-reinforcement. We see an almost *oppositve* behaviour.

Prior work on LLMs-as-facilitators indicates that they are way too overeager to intervene [CITE]. Un-

Table 3: Classifier performance trained on all datasets

dataset	precision	recall	f1	support
ceri	0.379	0.122	0.185	90
fora	0.407	0.624	0.493	1345
iq2	0.452	0.568	0.504	1264
umod	0.316	0.300	0.308	20
whow	0.420	0.584	0.489	890
wikitactics	0.576	0.292	0.388	130

Table 4: Classifier performance trained on written datasets

dataset	precision	recall	f1	support
ceri	0.350	0.544	0.426	90
umod	0.212	0.438	0.286	16
wikitactics	0.371	0.705	0.486	122

Table 5: Classifier performance trained on spoken datasets

dataset	precision	recall	f1	support
fora	0.422	0.600	0.496	1341
iq2	0.430	0.621	0.508	1228
whow	0.412	0.591	0.486	942

like their setup, which simply prompted LLMs to intervene "when absolutely necessary", our setup explicitly asks the models whether they would have intervened with positive or negative reinforcement, or without intervention at all. However, the rates of interventions seem to be the same as the ones reported by the authors, suggesting that the *over-eagerness to intervene cannot be solved by prompting*. Therefore, we believe that out-of-the-box LLMs can not be used for facilitation.

5 Can we predict when a facilitator would intervene?

As discussed in §2.1, facilitation interventions can be defined in two ways; whether a comment is facilitative in nature, or whether a dedicated facilitator would participate at a given point in time. So far we have tried to answer the former, which is only achievable through human annotation (Falk et al., 2024).

The latter definition may be less useful for analysis, but is much easier to answer (which is one of the reasons most relevant work uses this definition implicitly–§2.1). Indeed, considering when a facilitator would speak in our dataset as a ground truth, we can transform the task into a classification problem, where we are looking for the last comment made by a non-facilitator before a facilitator spoke.

6 Conclusion

Humans are good but expensive. Our research is good. We wasted a good amount of money on this, please accept this.

7 Limitations

None, we are awesome.

8 Ethical Considerations

All experiments and data have passed through official ethical approval by an Ethics Committee, and are compatible with GDPR and the EU AI Act. The annotators signed informed consent forms before beginning annotation and were allowed to skip uncomfortable tasks at any point, with no required justification.

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A Dataset details

A.1 Preprocessing

General

- We exclude comments with no text.
- We exclude discussions with less than two distinct participants.
- We exclude discussions which are common between wikitactics and wikiconv as well as wikidisputes and wikiconv.
  - There may be duplicate discussions between wikidisputes and wikitactics, but we allow them since they feature complementary information.

Wikiconv

- The Wikiconv corpus does not contain information about which user is a moderator/facilitator. Therefore, all comments relating to Wikiconv are tagged as non-moderators.
- Additionally, we follow the instructions of the original researchers (Hua et al., 2018), and select only discussions which have at least two comments by different users.

- Wikipedia (thankfully) does not track users who log in with only an IP address (in the original dataset, their `user_id` is always set to 0 and their username is of the form `211.111.111.xxx`. We consider each such username to be a separate user.
- Due to the size of the dataset, we have to partially load it during preprocessing. Thus, there is a small chance every 100,000 records that a discussion is marked as a false negative and a part of it gets discarded.
- We only include English comments in the final dataset. We use a small, efficient library (`py3langid`) for language recognition, due to the large size of Wikiconv. Non-English comments are discarded before selecting valid discussions (see point above).

Wikitactics

- We infer facilitative actions by whether the comment belongs in any of the following categories:
  - Asking questions
  - Coordinating edits
  - Providing clarification
  - Suggesting a compromise
  - Contextualisation
- The above tactics are a subset of the Coordinative labels used in the WikiTactics paper. They were selected because they are not used necessarily on 1-1 discussions; they could reasonably be applied by third-party participants. Contrast them with other Coordinative labels such as "Conceding/recanting" and "I don't know".

Wikidisputes

- Since only 0.03% of the comments in the dataset are made by moderators, we mark the dataset as not supporting moderation.

UMOD

- Facilitative actions are marked as a gradient from 0 (no facilitation) to 1 (full facilitation). We adopt a threshold of 0.75 to consider an action as facilitative, with more than 50% annotator agreement (measured as entropy in the original dataset).

CMV-AWRY2

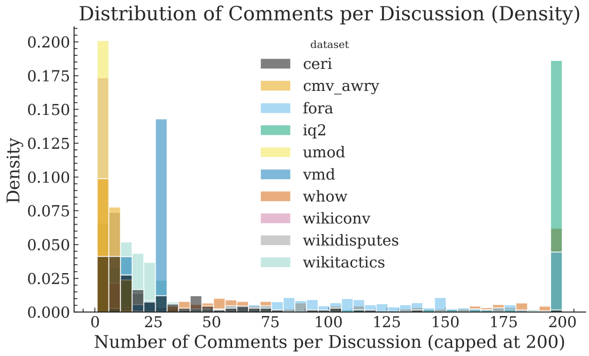

Figure 6: Comments per discussion on the PEFK dataset. Discussions over 40 comments have been clipped at the end of the x-axis for the purposes of graph readability.

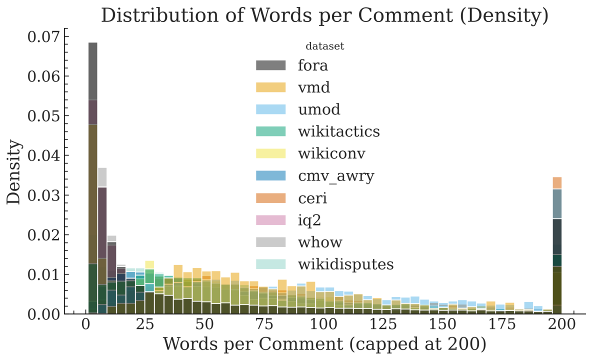

Figure 7: Words per comment on the PEFK dataset. Comments with 500 words have been clipped at the end of the x-axis for the purposes of graph readability.

- We mark a discussion as escalated when the derailment value (from the official dataset) is in the 60th upper percentile.
- We remove deleted ("[deleted]") comments.

A.2 Schema

Table 6 details the information available for each comment in the dataset. Some information, such as whether the person talking is a moderator / facilitator, is available only on some sub-datasets. Also note that the `is_moderator` column may hold different definitions of what we consider as moderator / facilitator. UMOD for instance considers facilitative comments made by users, while others such as Fora consider a comment facilitative if it was made by a dedicated facilitator.

A.3 Statistics

Figures 6,7 show the number of comments in each discussion, and the number of words per comment in our dataset respectively. We can see that most discussions are comprised of only a few comments, although there is a significant tail of very long dis-

Name	Type	Description
conv_id	string	The discussion's ID. Comments under the same discussion refer to the same discussion ID.
message_id	string	The message's (comment's) unique ID.
reply_to	string	The ID of the comment which the current comment responds to. nan if the comment does not respond to another comment (e.g., it's the Original Post (OP)).
user	string	Username or hash of the user that posted the comment
is_moderator	bool	Whether the user is a moderator/facilitator.
moderation_supported	bool	True if the moderation labels are directly computed from the original dataset
escalated	bool	A discussion-level measure denoting discussions which have been derailed
escalation_supported	bool	True if the escalation labels are directly computed from the original dataset
text	string	The contents of the comment
dataset	string	The dataset from which this comment originated from
notes	JSON	A dictionary holding notable dataset-specific information
toxicity	float	The "toxicity" score given to the comment by the Perspective API
severe_toxicity	float	The "severe toxicity" score given to the comment by the Perspective API

Table 6: The full set of columns used for each comment in the PEFK dataset.

cussions. Similarly, most comments are relatively short (<100 words), although there is again a long tail of very long comments.